USN | | | | | | | | | | [22TAIDEDE12]

FIFTH SEMESTER B.Tech EXAMINATION, NOVEMBER-2024

DOMAIN SPECIFIC AI

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART-A	**5x2=10**		
1.	**a.**	Illustrate the importance of domain specific AI with an appropriate example.	L2	1	**02**
	b.	Explain about hyperparameter tuning process with its key settings.	L2	2	**02**
	c.	Explain the key aspects of named entity recognition with its important features.	L2	3	**02**
	d.	Outline the various feature extraction techniques used in structured machine learning data.	L2	4	**02**
	e.	Show the key features of Explainable AI that focuses on making AI models.	L2	5	**02**
		PART-B	**6X5=30**		
2.	**a.**	Identify the key components of AI with an appropriate example.	L3	1	**05**
		OR			
	b.	Outline the future of AI techniques with its details.	L3	1	**05**
3.	**a.**	Identify the various applications of AI with an appropriate example.	L3	1	**05**
		OR			
	b.	Summarize the types of AI and its key aspects.	L3	1	**05**
4.	**a.**	Make use of clustering-based unsupervised learning with K-Means and hierarchical clustering approach.	L3	2	**05**
		OR			
	b.	Apply linear Regression model for some applications with its purpose and methods.	L3	2	**05**
5.	**a.**	Explain the basics of natural language processing with text preprocessing.	L3	3	**05**
		OR			
	b.	Illustrate any practical applications of NLP in healthcare domain.	L3	3	**05**
6.	**a.**	Show some real-time applications of computer vision in detail.	L3	4	**05**
		OR			
	b.	Explain the steps involved in the image classification process in detail.	L3	4	**05**
7.	**a.**	Explain the key concepts of transfer learning in detail.	L3	5	**5**

		OR			
	b.	Outline the benefits and challenges of transfer learning.	L3	5	**05**
		PART-C	**6 X 10=60**		
8.	**a.**	Identify the various evolution of domain specific AI in detail.	L3	1	**10**
		OR			
	b.	Show the types of supervised learning techniques and its purpose.	L3	1	**10**
9.	**a.**	Identify the various applications of AI with an appropriate example.	L3	1	**10**
		OR			
	b.	Identify different aspects to compare Narrow AI and General AI.	L3	1	**10**
10.	**a.**	Examine the process of hyperparameter tuning for domain-specific tasks with appropriate example.	L4	2	**10**
		OR			
	b.	Examine the process of optimal behavior learning using Reinforcement Learning concepts.	L4	2	**10**
11.	**a.**	Discover the steps involved in the sentiment analysis process with example.	L4	3	**10**
		OR			
	b.	List the types of tokenization used in NLP and its approaches with example.	L4	3	**10**
12.	**a.**	Explain the key concepts of Computer Vision with appropriate examples.	L4	4	**10**
		OR			
	b.	Analyze the practical applications of computer vision in manufacturing industry with its steps.	L4	4	**10**
13.	**a.**	Discover the steps involved in transfer learning for domain-specific AI context.	L4	5	**10**
		OR			
	b.	Discover any five important applications of reinforcement learning and explain in detail.	L4	5	**10**